Riley Beenders

PRODUCT MANAGEMENT | DATA PRODUCTS & PIPELINES | SYSTEMS IMPLEMENTATION

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PROFESSIONAL SUMMARY

Engineer who owns products from discovery through launch and builds the data systems that show a business what is actually happening, from a plant-wide production data service to an ERP rollout and a commercial product line. Breaks large work into shippable milestones, works across engineering, operations, vendors, and leadership, and reports status, risks, and budgets plainly.

CORE EXPERTISE

Product Management: Discovery through launch; product requirements for engineering and external partners; MVP scoping and iteration; roadmap milestones and dependency tracking; risk analysis and mitigation; commercialization against unit goals; project budgets, BOMs, and unit cost; stakeholder reviews with global leadership; Kanban backlogs in Microsoft Teams

Data & Pipelines: Data pipeline design; data migration and reconciliation; master data (items, products, SKUs); SQLite and SQL; structured run records and CSV exports; real-time data streaming with Server-Sent Events; production statistics dashboards; data logging and process analysis; Modbus TCP data acquisition

Platforms & Tools: Microsoft Dynamics 365 Business Central (SaaS on Azure); SimplyPrint; Python (asyncio, Starlette); TypeScript, React, and Next.js; Kubernetes and Docker; Git and GitHub; GitHub Actions; Windows/Linux systems administration

Operations & Communication: Root cause analysis; incident reports for leadership; operational documentation and user guides; technical writing; vendor coordination; worldwide project collaboration; hiring manager and technical interviews; Lean Six Sigma

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA — January 2025 - Present

Lead, Research & Development Engineer

Proteor acquired Filament Innovations in January 2025, and the CTO role below continued as Lead R&D Engineer for the same product line, now Proteor Print.

- Took the Coplay location's Microsoft Dynamics 365 Business Central, Microsoft's SaaS ERP hosted on Azure, from evaluation to live production in five weeks: two weeks evaluating the software, then three migrating more than 2,000 items and 150+ products into the live system, reconciling inventory counts on the floor, and syncing SKUs with Proteor's global locations. The whole facility of 10 to 15 people works in it, and 10 to 30 more at Proteor locations worldwide reference it.
- Traced the discrepancies in the 2026 Business Central inventory count, which I led across every item, to production orders processed without verification that never took items out of inventory, and helped move the plant to a more warehouse-focused workflow.
- Took ExtrusionStudio, the plant's production data service, from a LabVIEW proof of concept to a Python service live on all three filament extrusion lines since October 2026, with no unplanned outage during production, so every spool's diameter is captured instead of written down by hand.
- Use ExtrusionStudio's production history to estimate how many spools of each material the plant can complete in a week, and keep a running history of the schedule and material parameters.
- Replaced full-sheet label printing with print-on-demand thermal labels, removing up to about 30% of label waste per run.
- Contributed the machine requirements and integration for ProteorPrintOS, built with an external software partner on SimplyPrint, giving users remote operation, live video, AI-assisted failure alerts, and material, waste, and outcome metrics.
- Owned the ICARUS-Lite, a compact industrial printer for prosthetic and orthotic production, from design and development through commercialization against unit goals of 60 in 2025 and 100 in 2026.
- Estimate and track the budget of every engineering project, build its BOMs, and maintain each product's unit cost in Business Central.
- Present project status, constraints, and budgets at Proteor's monthly USA review with leadership from its global locations, and write incident reports for leadership.
- Co-developed a custom hotend with an outside partner, improving reliability and reducing cost for our materials.
- Developing an offline version of the ICARUS-Lite and ICARUS-Pro control screen with over-the-air automated updates, so customers can run without a Wi-Fi or Ethernet connection.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.

- Serve as the plant's technical subject-matter expert and the escalation point for customer issues that go beyond what support can resolve, across mechanical faults, electrical faults, and user error.
- Work directly with equipment and software vendors to resolve issues on the Siemens PLC-controlled extrusion lines and the Duet3D printer controls, regularly into nights and early mornings.
- Managed and expanded the division's IT infrastructure after the acquisition, standardizing software and internal processes to align with Proteor's company-wide systems.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.

Filament Innovations | Whitehall, PA — May 2020 - January 2025

Across these roles, wrote customer proposals, scopes of work, and estimates, and began tracking project budgets.

Chief Technology Officer — January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while keeping hands-on responsibility for mechanical, electrical, firmware, and software systems, and gave the CEO and the company regular updates as designs and software iterated.
- Led the design and launch of Gen3 POSEIDON, the flagship's production redesign, combining a 1 m³ build volume, filament-and-pellet dual extrusion, and braked Z drives that prevent uncontrolled motion during power loss.

Lead Electro-Mechanical Engineer — June 2022 - January 2023

- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.
- Removed a manufacturing bottleneck with independently testable electronics enclosures that installed in minutes, coordinating custom harnesses, cut panels, and hardware improvements with vendors.
- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging on larger, modular machines.

Industrial Engineer — April 2021 - June 2022

- Delivered THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry machine for a client producing the world's largest 3D-printed statue, carrying it from CAD, BOM, and sourcing through fabrication, assembly, and firmware tuning to customer support and operator training.
- Designed and released Kratos, a machine engineered to print high-strength above-knee and below-knee prosthetic sockets.

Engineering Technician — May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized machine parameters while documenting design limitations and improvement opportunities.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed more than 750 wireless access points across six school buildings; accelerated deployment of more than 2,500 Chromebooks with an Arduino-based tool that automated setup-data entry, and coordinated final device distribution with school administrators.

Blue Mountain Ski Resort - Ski Lift Operator (seasonal) | November 2019 - January 2021 | Operated lifts and assisted thousands of guests with ticketing, weather, and slope information; performed motor and control safety checks and supported the maintenance team during lift breakdowns.

Carver House Homeowners Association - Treasurer | April 2026 - Present | Manage the banking and monthly insurance payments for an association of about 80 units, and help establish a community-focused member board in place of a property management company, remaining Treasurer through a mid-year restructure.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering — December 2024

Machine Learning Onramp, MathWorks	2026
MATLAB Onramp, MathWorks	2026
Simulink Onramp, MathWorks	2026
Simscape Onramp, MathWorks	2026
Lean Six Sigma Yellow Belt, Educate 360	2025
Lean Six Sigma White Belt, Educate 360	2025
Certificate in Engineering Design and Tools, Penn State University	2023
Mechanical Engineering Certificate, Lafayette College	2019

SELECTED PROJECTS

ExtrusionStudio | Plant measurement and production data system, Proteor — September 2025 - Present

Python, asyncio, Modbus TCP, Starlette, SQLite, Server-Sent Events, Raspberry Pi

Problem: Each of three extrusion lines had a tri-axial laser gauge, and the only record of a spool's diameter was whatever an operator wrote down. Production was speeding up, and the plant needed every reading captured, every spool judged against its tolerance, and the result visible to the operator, the packing bench, and management at once.

- Runs one asyncio poller and Modbus TCP session per line, reading every line concurrently at 20 Hz so a powered-down line never stalls a healthy one; the poll rate is per-line configuration with a floor, because the right value depends on the PLC's response time.
- Added a calibration phase that times the filament from the laser to the spool, a delay of one to two minutes through the accumulator, so each reading and alarm lands on the spool it is actually wound on rather than the one under the laser.
- Logs every spool to its own CSV and a SQLite run table with a serial; the packing bench approves, inspects, or rejects each spool with the reasons recorded on its serial, and an approved spool gets an HTML certificate.
- Split the interface into a view per audience (a projector view for the floor, a touch screen per line, the packing bench, and a production statistics TV for management), streamed live over Server-Sent Events.

Engineering Operations Platform | Internal ERP/CRM, Filament Innovations (as CTO)

Kubernetes, ERP/CRM development, BOM automation, infrastructure design

Problem: Orders, inventory, tasks, projects, customer interactions, engineering BOMs, and purchasing ran as separate manual workflows.

- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM that centralized those records, generated BOMs dynamically, and linked design data directly to purchasing, reducing the separation between design decisions and purchasing execution.

RileyBeenders.com | Data-driven portfolio site, rileybeenders.com/about-this-site — June 2026 - Present

Next.js, React, TypeScript, Vercel, GitHub Actions, Playwright, Claude Code

Problem: Content, layout, and styling started tangled in one file, so every visual change risked the content and every content change risked the layout.

- Split one resume JSON into focused data files (header, summary, experience, skills, education, projects) merged at build time, with one source feeding the site, its job-application tracker, and every tailored resume.
- Content changes from a local schema-driven Studio without touching code, and a GitHub Actions workflow refreshes the site's statistics and timeline from git on every push to main, so the site's own record stays current without manual upkeep.

Unified UniFi Network | Network infrastructure, Proteor plus two residential sites — March 2024 - Present

Ubiquiti UniFi cloud gateways, VLANs, DHCP, DNS, network monitoring

Problem: As production grew, the plant's consumer mesh network assigned the same IP address to machines that came online together and never reliably detected or resolved the conflicts.

- Replaced it with a centrally managed UniFi network, using VLANs to separate production equipment, work devices, and IoT devices, which reduced IP conflicts and network-related downtime; all three sites are configured and monitored remotely from one interface.

BulkSSH | Windows desktop app, github.com/RileyBeenders/BulkSSH — October 2026 - Present

TypeScript, Electron, Node.js, SSH

Problem: Testing ExtrusionStudio's Pi kiosks meant repeating the same commands on every Pi, one SSH window at a time, after every change to the image.

- Opens many devices in a grid of live terminals, mirrors keystrokes to every ticked device, or broadcasts one command and collects each device's output and exit code, so an image change is tried on every Pi in one pass; released as an installer and a portable executable.

RB's Auto App Updater | Windows utility — September 2026

PowerShell, WinGet, Windows Task Scheduler, GitHub, SHA-256 verification

Problem: Windows applications update through different methods, so keeping dozens of them current meant visiting websites, downloading installers, and running each one by hand.

- Updates every WinGet-managed application with one double-click and no UAC prompt after a one-time setup, upgrading each package on its own so a single failure never stops the rest, and translates WinGet exit codes into plain-language statuses such as security blocked, restart needed, and application in use.